president@dnsc.edu.ph
dnsc.edu.ph
@officialdnsc
New Visayas, Panabo City, 8105

Institute of Computing

Progress Report

SKILLBRIDGE: A WEB-BASED OJT PLACEMENT DECISION SUPPORT SYSTEM WITH NLP-ASSISTED SKILLS ASSESSMENT AND COMPANY-POSITION RECOMMENDATION

Researchers: David Rey P. Bali-os, Lemuel P. Brion and Azel M. Villanueva

As of: October 20, 2026

Pre-deployment defense: Informal: November 11-13, 2026 | Formal: November 16-17, 2026

Introduction

SkillBridge supports OJT placement decisions at Davao del Norte State College through student records, competency assessments, company-position requirements, NLP-assisted matching and analytics. This report presents progress since the June 2, 2026 outline defense. The system is in Implementation, with working core features and continuing UI refinement and local verification.

Updated Gantt Chart

The revised schedule runs from planning in early 2026 to maintenance in June 2027. Implementation extends through January 2027, including system deployment in January. Maintenance runs approximately February-June 2027, with UAT and evaluation in February-March. The November informal and formal defenses are separate review milestones.



The team-supplied chart is reproduced unchanged. A larger landscape copy appears at the end. Month ranges are approximate readings of the quarter-based image; exact day-level dates are not inferred.

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Project Status

SkillBridge is a working prototype in the Implementation phase. Core workflows are implemented locally; selected UI improvements, regression verification and manuscript alignment continue. Hosted deployment and formal user evaluation remain upcoming activities.

Recent Accomplishments

- **Manage project information.** Implemented student profiles, company and position records, skill requirements, competency results, recommendations and placement records, with separate role access. Batch management, instructor assignment and IAMS-format enrollment support record preparation.
- **Provide competency assessments.** Implemented authoring and import, Multiple Choice, True/False and Identification questions, randomized delivery, automatic scoring and category results. Timed attempts, authorized retakes and combined competency profiles support the assessment workflow.
- **Compare NLP preprocessors.** Developed comparison tools for pretrained spaCy Small, spaCy Medium and Stanza English pipelines, with recommendation metrics, timing records and explicit reporting of unavailable models. Final interpretation and manuscript alignment continue.
- **Generate hybrid recommendations.** Implemented ranking using category, NLP-assisted text and location similarity with 60%, 25% and 15% weights. Completion, capacity and staff approval remain separate checks. Updated safeguards block outdated matches until refreshed.
- **Provide analytics and reports.** Implemented dashboards and reports for assessment completion, competency results, recommendations, placements and remaining slots, with clearer student tables, chart drill-downs and scoped exports.

Challenges and Solutions

Development and internal checking identified the following issues and responses.

- **Accurate assessment scores.** Unanswered questions and invalid submissions required more consistent handling. We revised answer validation and full-question category totals, with checks for incomplete answers, retakes and repeated submissions.
- **Slow or uncertain submissions.** Model loading and recommendation processing could delay feedback. We extended the submission wait and check the recorded attempt after failed responses, recognizing saved results while retaining unconfirmed answers. Performance refinement continues.
- **Recommendations after record changes.** Updated position requirements could leave older matches available. We added matching-input checks, blocked outdated results and provided an explicit refresh while preserving placement history.
- **Clear staff responsibilities.** Assessment and batch workflows needed clearer ownership and administrator controls. We added shared authoring, instructor assignment, scoped management and action history while enforcing role boundaries.
- **Consistent result presentation.** Several pages needed clearer grouping, navigation and feedback. We improved dashboards, report views, company details and shared notifications. Selected page refinements remain ongoing.

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Comparison of Actual Progress to the Planned Timeline

The revised Gantt chart provides the phase windows below. Earlier planning, analysis and design outputs are available, but their exact actual completion dates were not independently established. Current code and local verification support the implementation status shown.

Phase or activity	Planned window	Progress as of October 20, 2026
Planning	Feb-Mar 2026	Scope, study objectives and project schedule documented.
Analysis	Mar-Apr 2026	OJT process, roles and requirements documented.
Design	Apr-May 2026	Architecture, UI and database/data-flow designs available; refinements continue.
Implementation	Jun 2026-Jan 2027	In progress. Core workflows implemented; local checks and UI refinement continue.
System Testing & QA	Sep-Oct 2026	In progress. Local checks have been conducted; remaining verification continues.
System Deployment	Jan 2027	Not yet started.
UAT & Evaluation	Feb-Mar 2027	Not yet started.
Maintenance	Feb-Jun 2027	Not yet started.

The frontend/backend development bars end around September 2026, while NLP integration and QA extend into October. Selected UI refinements are still ongoing after the core-development window and are not shown as a separate bar. Exact schedule variance is not calculated from the image alone.

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Work Completed

- **Data and account management:** Built role-based workflows for student records, batches, companies, positions, requirements and OJT placement records.
- **Assessment workflow:** Built question authoring and import, assessment deployment, timed answering, scoring, category results and instructor-controlled retakes.
- **Combined competency evidence:** Added batch-scoped competency profiles and required-assessment completion checks for recommendation access.
- **NLP comparison tools:** Integrated pretrained preprocessing pipelines with model comparison and diagnostics. TF-IDF/cosine similarity supports text matching; diagnostics identify the actual model and unavailable runs. The application does not train a new language model.
- **Recommendations and placement review:** Built ranked matches, separate score components, staff approval controls, capacity checks and saved placement snapshots.
- **Analytics and report generation:** Built scoped dashboards, assessment and placement summaries, student detail views and downloadable reports. Recent local work improves table consistency and interactive charts.
- **Verification and readiness records:** Prepared local automated and browser checks, an independent code review, a response matrix for the 17 panel comments, and release/recovery guidance.

The following pages contain archived sample UI screens, a source-code excerpt and a generated report. The team will replace this sample evidence with current localhost captures and corresponding captions before submission, using the separate evidence capture guide.

UI and report examples were captured locally on October 17-18 using synthetic records; the code excerpt was prepared on October 20. The figures show implementation, rather than hosted acceptance or actual placement outcomes.

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Evidence of Work: Code, Interfaces and Generated Output

- **Company and position management:** The interface shows a position, available slots and required skill levels for placement preparation.

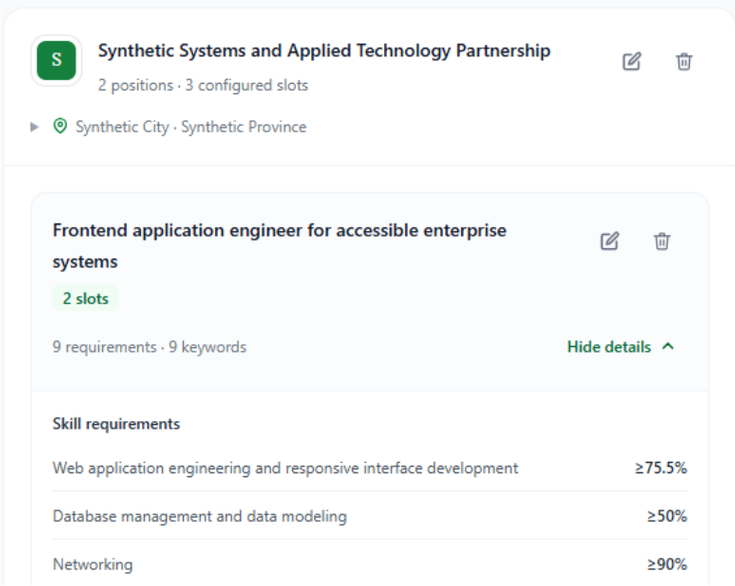


Figure 1. Companies and Positions interface with a synthetic company record, position, slots and requirements. Local UI fixture, October 17.

- **Assessment authoring:** The editor provides question types, answer fields and skill-category controls.

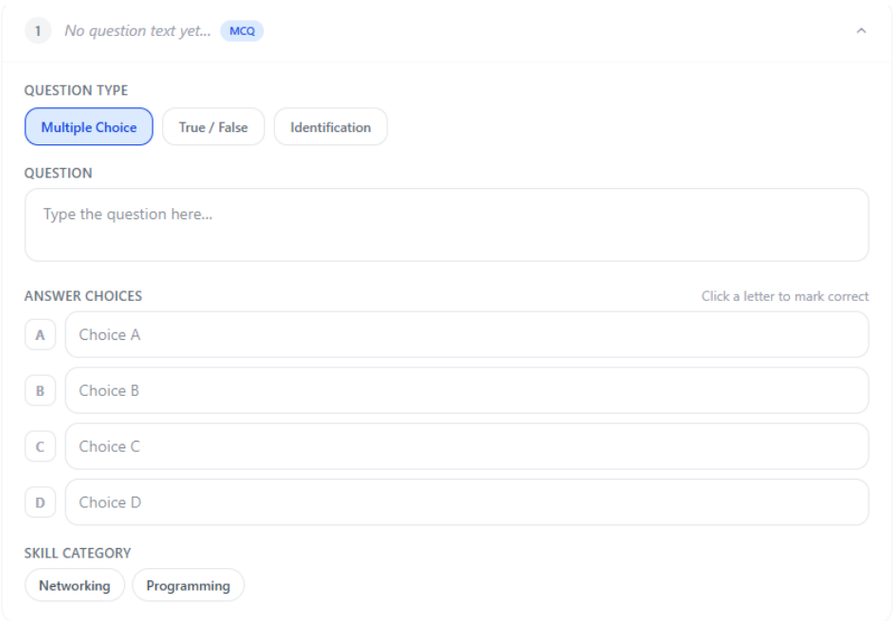


Figure 2. Assessment authoring controls from the local prototype, October 18. Scoring verification is recorded separately.

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Evidence of Work: NLP Comparison and Recommendations

- **NLP comparison:** The model registry and evaluation loop configure three preprocessors and check the actual model used during strict comparison. Source filenames appear in the excerpt.

api/recommendation_nlp.py

```
20  MODEL_OPTIONS = {
21      'spacy_sm': {
22          'label': 'spaCy Small',
23          'package': 'en_core_web_sm',
24      },
25      'spacy_md': {
26          'label': 'spaCy Medium',
27          'package': 'en_core_web_md',
28      },
29      'stanza_en': {
30          'label': 'Stanza English',
31          'package': 'stanza English resources',
32      },
33  }
```

api/nlp_evaluation.py

```
193      for model_id, info in MODEL_OPTIONS.items():
194          row = {'id': model_id, **info, 'status': 'unavailable', 'metrics': None}
195          try:
196              _load_model(model_id)  # Warm-up excluded, matching notebooks. Never
     download.
197          except Exception:
     ...
203              start = perf_counter()
204              processed, actual, reason = preprocess_texts(texts, model_id,
     allow_fallback=False)
205              if actual != model_id or reason:
206                  raise ValueError('Evaluation requires the requested model.')
```

Figure 3. Labeled source-code excerpts prepared October 20. This shows comparison implementation; experiment results and final model selection require separate explanation.

- **Ranked suggestions:** Staff can inspect a student's suggested position, total match, category/NLP/location components, distance and available capacity before making a placement decision.

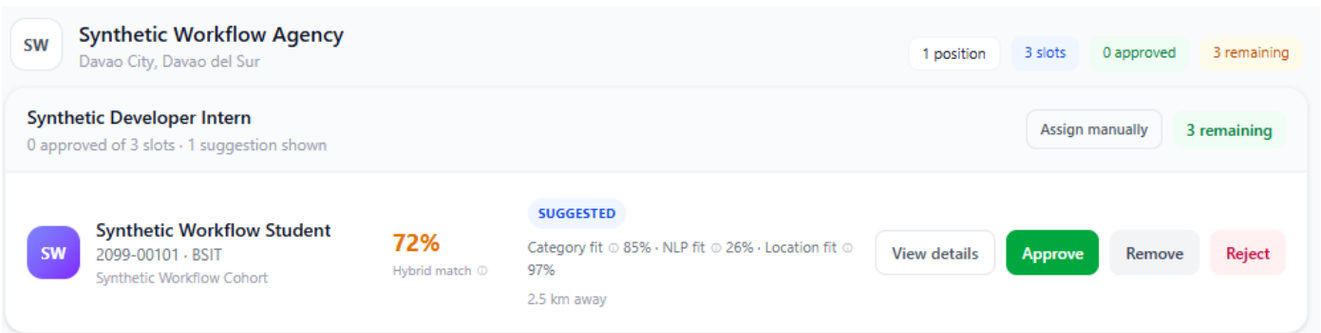
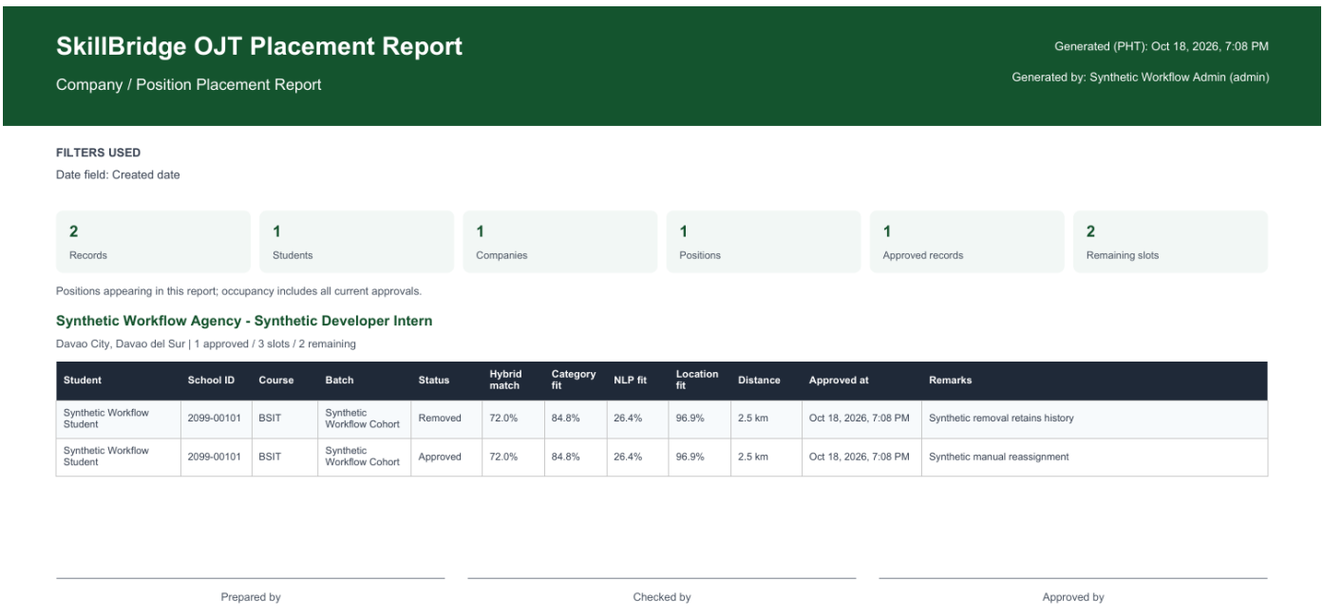


Figure 4. Synthetic company-position suggestion from a local React/API demonstration, October 18. Match values are illustrative; approval remains a separate staff action.

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Evidence of Work: Analytics and Report Generation

- **Placement reporting:** The company placement workflow produces a PDF containing placement history, match components and remaining capacity.



SkillBridge OJT Placement Report

Company / Position Placement Report

Generated (PHT): Oct 18, 2026, 7:08 PM

Generated by: Synthetic Workflow Admin (admin)

FILTERS USED

Date field: Created date

2	1	1	1	1	2
Records	Students	Companies	Positions	Approved records	Remaining slots

Positions appearing in this report; occupancy includes all current approvals.

Synthetic Workflow Agency - Synthetic Developer Intern

Davao City, Davao del Sur | 1 approved / 3 slots / 2 remaining

Student	School ID	Course	Batch	Status	Hybrid match	Category fit	NLP fit	Location fit	Distance	Approved at	Remarks
Synthetic Workflow Student	2099-00101	BSIT	Synthetic Workflow Cohort	Removed	72.0%	84.8%	26.4%	96.9%	2.5 km	Oct 18, 2026, 7:08 PM	Synthetic removal retains history
Synthetic Workflow Student	2099-00101	BSIT	Synthetic Workflow Cohort	Approved	72.0%	84.8%	26.4%	96.9%	2.5 km	Oct 18, 2026, 7:08 PM	Synthetic manual reassignment

Prepared by

Checked by

Approved by

Figure 5. Detail of a report downloaded from the local workflow on October 18. Synthetic Removed and Approved entries illustrate retained history and capacity reporting. Blank signature fields do not represent approval or endorsement.

Supporting Verification

The latest source rescan covers 345 application, configuration, migration and test files, with focused tracing of updated assessment, recommendation, management and analytics workflows. Fresh frontend lint/build checks and 135 unit tests passed. The broad isolated SQLite backend run discovered 347 tests: 335 passed and 12 PostgreSQL-specific tests were skipped. These are implementation checks, not participant-evaluation or model-effectiveness results.

Earlier local browser and PostgreSQL records remain supporting evidence; they were not rerun in the latest rescan. The review used disposable test settings with cache isolation and did not operate the live database or hosted services. Final intended-environment verification remains pending.

The screenshots and generated report above use synthetic local examples captured October 17-18. They demonstrate selected implemented features and do not show every subsequent UI refinement or actual deployment outcomes.

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Work in Progress

Selected UI refinements, regression checks, manuscript alignment and evidence capture continue. We target adviser review October 26-31 and fixes/retesting November 1-7. The latest formal panel record is the June 2 outline-defense minutes.

Anticipated Completion and Activity Windows

Task	Planned timing
Adviser review of current system and manuscript	October 26-31, 2026 (target)
Apply adviser feedback, fix issues and retest revisions	November 1-7, 2026 (target)
Informal pre-deployment defense	November 11-13, 2026
Formal pre-deployment defense	November 16-17, 2026
UI refinement, evidence capture and manuscript alignment	Ongoing during Implementation
System deployment and intended-environment verification	January 2027
UAT and formal evaluation	Approximately February-March 2027
Bug fixes, security activities and system monitoring	Approximately February-June 2027

Adviser review dates are targets, not a confirmed appointment. The defense schedule supplements the Gantt; local checks remain separate from formal UAT/evaluation.

Conclusion

Key Findings and Achievements

SkillBridge implements project records, assessments, NLP comparison tools, hybrid recommendations and analytics/reports. The figures illustrate progress since the outline defense.

Assessment of Project Progress Against Goals

The prototype addresses the five study objectives, with UI refinement and verification continuing within Implementation. Readiness for actual use still requires intended-environment checks, genuine user evaluation and adviser review.

Plans and Next Steps

- **Late October-early November:** Seek adviser review October 26-31 and apply and verify requested revisions November 1-7.
- **November defenses:** Present at the November 11-13 informal defense, address feedback and prepare for the November 16-17 formal defense.
- **January-June 2027:** Prepare for January deployment, followed by scheduled UAT/evaluation and maintenance through June.

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SkillBridge Progress Report - Updated Project Gantt Chart

Team-supplied schedule, 20 October 2026. Month ranges in the report are approximate readings of this chart.

